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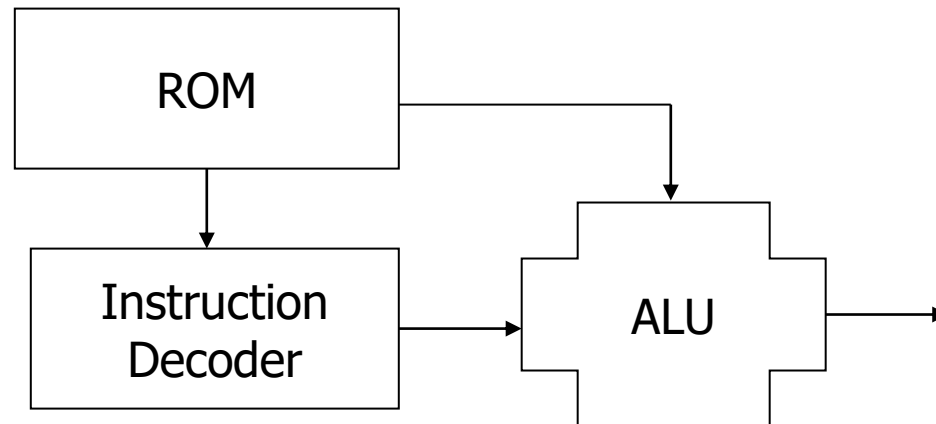
WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 1F: Component Instantiation

Structural VHDL: Component Instantiation

- Component Instantiation
 - Helps break down a design into subsections for ease of understanding and design reuse

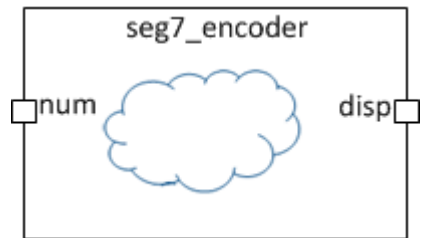


Structural VHDL: seg7_encoder

- Given the component **seg7_encoder**, we want to use this component twice
 - How do we put it down and hook it up?

- Note:

- The entity and architecture for seg7_encoder are in another file



architecture structure of some_top_level_design is

```
signal disp1_data, disp2_data : std_logic_vector (3 downto 0);
signal to_display_1, to_display_2 : std_logic_vector (6 downto 0);
```

```
begin
```

```
    enc1 : entity work.seg7_encoder port map (disp1_data,to_display_1);
    enc2 : entity work.seg7_encoder port map (
        num => disp2_data,
        disp => to_display_2);
end structure;
```

↑
Binding in
either way

- This form is called **direct entity instantiation** as it has the entity keyword in the instantiation, another form called component instantiation is shown on the next slide.
- Important to include either work.seg7_encoder in the instantiation or use work.all

Component Instantiation

- As an alternative to the recommended entity instantiation shown on the previous slide, you can use component instantiation, which pre-declares the component prior to instantiation.
 - In this case, the entity keyword is removed from the instantiation line

```
architecture structure of some_top_level_design is

    component seg7_encoder
        port map( num:  std_logic_vector(3 downto 0);
                  disp: std_logic_vector(6 downto 0));
    end component;

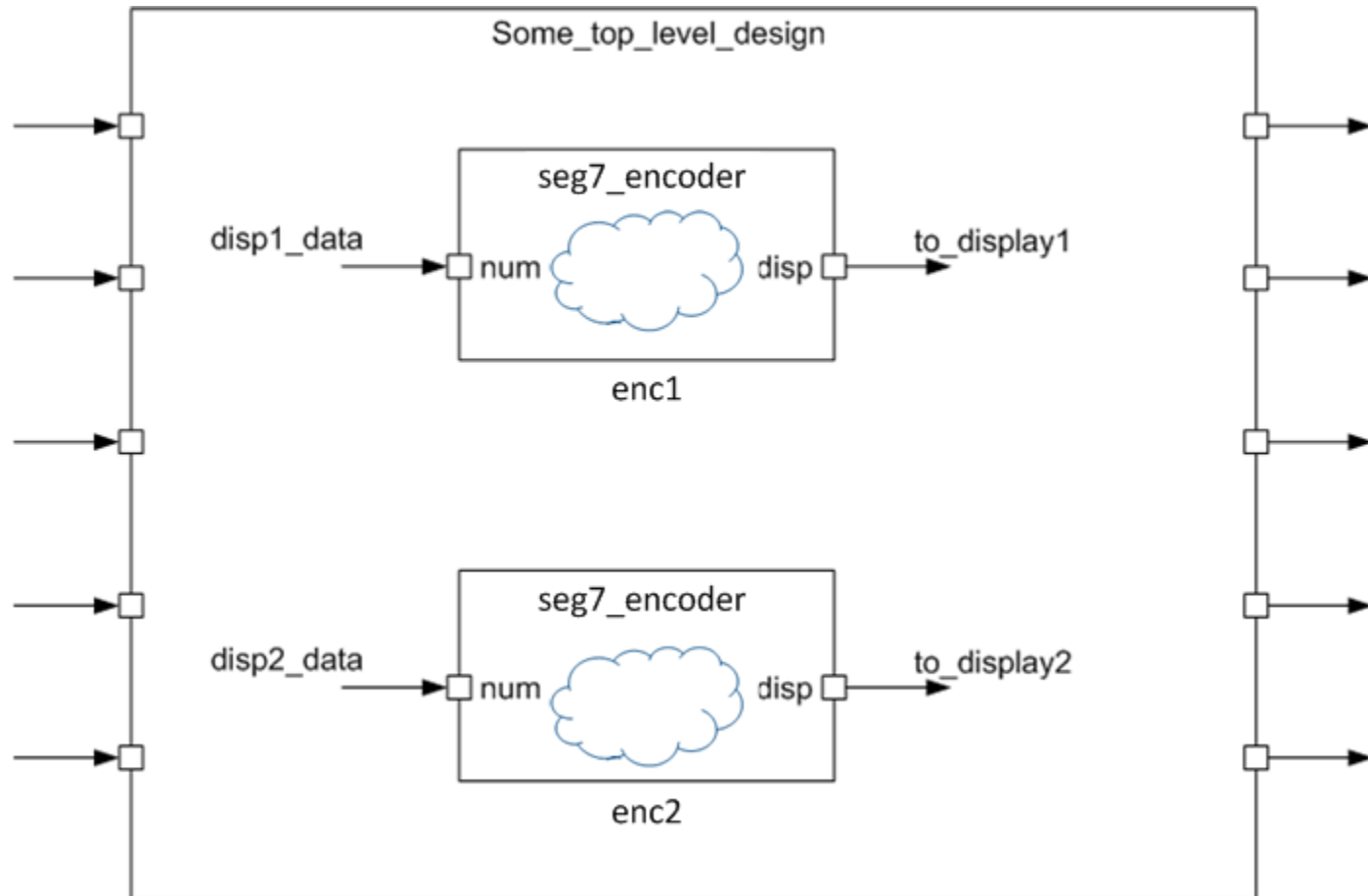
    signal disp1_data, disp2_data : std_logic_vector (3 downto 0);
    signal to_display_1, to_display_2 : std_logic_vector (6 downto 0);

begin

    enc1 : seg7_encoder port map (disp1_data,to_display_1);
    enc2 : seg7_encoder port map ( num => disp2_data,
                                disp => to_display_2);

end structure;
```

Structural VHDL



Component vs. Instance

architecture structure **of** some_top_level_design **is**

```
signal disp1_data, disp2_data : std_logic_vector (3 downto 0);  
signal to_display_1, to_display_2 : std_logic_vector (6 downto 0);
```

begin

Component name



```
enc1 : entity seg7_encoder port map (disp1_data,to_display_1);
```

```
enc2 : entity seg7_encoder port map (
```

```
    num => disp2_data,
```

```
    disp => to_display_2);
```



```
end structure; Instance name
```